

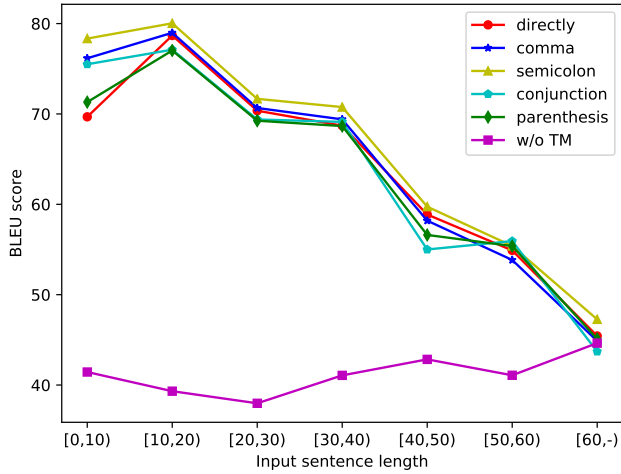


Figure 4: input sentence length impact on BLEU.

tence is retrieved. After calculation, we find that the average sentence length of the UNPC En-Zh test set and homemade En-Zh test set in Table 4 are 29.58 and 10.99. This is why the latter can improve the translation more significantly than the former even though there are fewer sentences with high similarity than there are in the former.

Related Work

Many studies have been conducted in recent years to enhance MT quality using TMs. With the emergence of NMT, the MT community is seeing an increasing interest in TM research. There are mainly two research lines for TM integration into NMT: constraining the decoding process with TM and using TM to train a more powerful NMT model.

The main idea of the first research line is to increase the generation probability of some target words based on TM. ? constrained the decoding process by increasing the generation possibility of the target words which are in the aligned slices extracted from the TM. Following this work, ? added positional information for words in the TM slices. ? and ? embedded the retrieved TM information into the NMT model by fine-tuning the NMT model with TMs before translating the input sentence. Instead of incorporating text level TM, k NN-MT retrieved TMs from dense vectors (?). First, they created a key-value datastore from the TM database, where the key is the translation context vector of each time step, and the value is the true target token. In the inference time, k NN-MT interpolates the generation probability of the NMT model and retrieved similar target distribution from that datastore at each time step. Following k NN-MT, several researches optimized k NN-MT from different perspectives. ? accelerated the inference process by narrowing the search range, instead of searching from the entire data store. By introducing a lightweight meta- k network, ? dynamically determines how many neighbors should be introduced. ? further accelerated k NN-MT inference by constraining search space when constructing the data store. Instead of retrieving a single token, ? retrieved chunks of to-

kens from the data store to speed up k NN-MT.

The second research line aims to train the generation model to learn how to deal with the retrieved TMs. ??? used a data augmentation way to concatenate the retrieved TM with input sentence during training. One sub-method of ? generated the target prefix in a forced way when decoding, while they need to train the NMT model with a similar concatenation pattern first. Some researches modified the NMT model architecture for better TM integration. ? and ? introduced a gating mechanism module to control the signal from the retrieved TM. ? designed an additional transformer encoder to integrate the target sentence of TM through the attention mechanism. In ?, the retrieved multiple TMs are compressed into a graph structure for speed up and space savings and then are integrated into the model via the attention mechanism. ? proposed a lightweight method to incorporate the target sentence of retrieved TM in an extra attention module. Unlike all of the above methods, ? proposed a method to incorporate monolingual TM into NMT, and the target sentence retriever and NMT model are trained jointly.

Conclusion and Future Work

In this paper, we propose a simple but effective method to incorporate TM into NMT decoding without modifying the pre-trained NMT model. Specifically, we treat the retrieved TMs as prompts for the translation of the input sentence by concatenating the source TM with the input sentence and generating the target token in a forced way. Experiments on the TM specialized translation task, domain adaptation task, and implementation on commercial MT system verify the effectiveness of our method. Our method is easy to implement and can be applied to customize a TM-incorporated machine translation system for TM data on the user side. Our method in this paper suffers from TM sentences with low similarity scores and long sentences. In the future, we will investigate more effective methods to alleviate the drawbacks of our methods in low similarity TM and long sentence translation situations.

Acknowledgments

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